Data Preprocessing - Theory

1. Missing Values

Missing values are observations where data is not available, usually represented by `NaN`. They can occur because of data-entry errors, incomplete information, or collection problems.

Simple Imputation

Replaces missing numerical values with a statistical value.

- **Mean:** Replaces missing values with the column mean.
- **Median:** Replaces missing values with the column median.
- **Most Frequent:** Replaces missing categorical values with the most common category.

Random Sample Imputation

Replaces a missing value with a randomly selected value from the observed values of the same column. It helps preserve the original distribution.

KNN Imputation

Uses the values of the **K nearest similar observations** to estimate a missing value. It considers relationships between variables.

Iterative Imputation / MICE

Predicts missing values using other variables and repeatedly updates the estimates until they become stable.

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2. Missing Indicator

A missing indicator is an additional variable that records whether the original value was missing.

- `0` → Value was present
- `1` → Value was missing

It helps preserve information about the original missingness after imputation.

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3. Outliers

An outlier is an unusually high or low observation compared with the rest of the data.

Example:

`18, 20, 21, 22, 200`

Here, `200` may be an outlier.

IQR Method

The Interquartile Range measures the spread of the middle 50% of the data.

`IQR = Q3 - Q1`

Outlier limits:

`Lower = Q1 - 1.5 × IQR`

`Upper = Q3 + 1.5 × IQR`

Values outside these limits are potential outliers.

Best suited for: skewed or non-normal data.

Z-Score Method

Z-score measures how far a value is from the mean in terms of standard deviations.

`Z = (x - μ) / σ`

A common rule is:

`|Z| > 3` → potential outlier.

Best suited for: approximately normally distributed data.

Percentile Method

This method uses lower and upper percentiles to identify extreme values.

For example:

- Lower limit = 5th percentile
- Upper limit = 95th percentile

Values outside these limits are treated as extreme observations.

Winsorization

Winsorization **caps extreme values instead of removing them**.

For example, if the upper limit is `50`:

`100 → 50`

The row remains in the dataset, but the extreme value is reduced.

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4. Outlier Detection vs Treatment

Detection identifies potential outliers using methods such as:

- IQR
- Z-score
- Percentile

Treatment decides what to do with them:

- Remove the outlier
- Winsorize/cap the value
- Keep it if it is a valid observation

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5. Project Approach

In this project, the techniques are applied **only to the specified columns**, not to every column.

The general process is:

Identify missing values → Apply selected imputation → Add missing indicator where required → Detect outliers → Remove or Winsorize outliers → Verify results.

Key Point

Different techniques are used for different columns depending on their data type and distribution. Therefore, a technique does not need to be applied to the entire dataset.